Brennan J. Sullivan, PhD

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Education

PhD, Neuroscience, Baylor College of Medicine 2021 - 2026
Dissertation: *Testing dendritic plateau potential-dependent plasticity in neocortex*
Advisor: Jeff Magee

BS, Neuroscience, Rhodes College 2013 - 2017

Research Experience

Postdoctoral Associate, Nuo Li Lab 2026 - Present
Duke University School of Medicine

PhD Student, Jeff Magee Lab 2022 - 2026
Baylor College of Medicine, Howard Hughes Medical Institute

Research Assistant, Shilpa Kadam Lab 2017 - 2021
Kennedy Krieger Institute, Johns Hopkins University School of Medicine

Undergraduate Research Fellow, Kelly Dougherty Lab 2016 - 2017
Rhodes College

Publications

Articles [* denotes equal contribution]

1. Xiao K.*, Li Y.*, **Sullivan B.J.***, Li G.*, Magee J.C. (2025) Rapid neocortical network modifications via dendritic plateau potential induced plasticity. *bioRxiv* 2025.11.19.689338.
2. Thomas A.*, Yang W.*, Wang C., Tipparaju S.L., Chen G., **Sullivan B.J.**, Swiekatowski K., Tatam M., Gerfen C., Li N. (2023) Superior colliculus cell types bidirectionally modulate choice activity in frontal cortex. *Nature Communications*. PMID: 37963894.
3. **Sullivan B.J.**, Kipnis P.K., Carter B.M., Kadam S.D. (2021) Targeting ischemia-induced KCC2 hypofunction rescues refractory neonatal seizures and mitigates epileptogenesis in a mouse model. *Science Signaling*. PMID: 34752143.
4. **Sullivan B.J.**, Ammanuel S., Kipnis P.K., Araki Y., Huganir R.L., Kadam S.D. (2020) Low-dose perampanel rescues cortical gamma dysregulation associated with parvalbumin interneuron GluA2 upregulation in epileptic Syngap1$^{+/-}$ mice. *Biological Psychiatry*. PMID: 32107006.
5. Kipnis P.K.*, **Sullivan B.J.***, Carter B.M., Kadam S.D. (2020) TrkB agonists prevent post-ischemic BDNF-TrkB mediated emergence of refractory neonatal seizures in CD-1 pups. *JCI Insight*. PMID: 32427585.
6. Kang J., Kadam S.D., Elmore J.S., **Sullivan B.J.***, Valentine H.*, Malla A.P., Grace A.A., Rahmim A., Loew L.M., Baumann M., Gjedde A., Boctor E.M., Wong D.F. (2020) Transcranial photoacoustic imaging of NMDA-evoked focal circuit dynamics in rat forebrain. *Journal of Neural Engineering*. PMID: 32084654.
7. Carter B.M., **Sullivan B.J.**, Landers J.R., Kadam S.D. (2018) Dose-dependent reversal of KCC2 hypofunction and phenobarbital-resistant neonatal seizures by ANA12. *Scientific Reports*. PMID: 30097625.

Reviews

1. **Sullivan B.J.**, Kadam S.D. (2021) Brain Derived Neurotrophic Factor in Neonatal Seizures. *Pediatric Neurology*. PMID: 33773288.
2. Kadam S.D., **Sullivan B.J.**, Goyal A., Blue M.E., Smith-Hicks C. (2020) Rett Syndrome and

CDKL5 Deficiency Disorder: From Bench to Clinic. *International Journal of Molecular Sciences*. PMID: 31618813.

3. Kipnis P.K., **Sullivan B.J.**, Kadam S.D. (2019) Sex Dependent Signaling Pathways Underlying Seizure Susceptibility and the Role of Chloride Cotransporters. *Cells*. PMID: 31085988.

Chapters

1. **Sullivan B.J.**, Kadam S.D. (2021) Protocol for Drug Screening with Quantitative Video Electroencephalography in a Translational Model of Refractory Neonatal Seizures. In *Experimental and Translational Models to Screen Drugs Effective Against Seizures and Epilepsy*. D. Vohora, Ed. Springer, New York.
2. **Sullivan B.J.**, Kadam S.D. (2020) The involvement of neuronal chloride transporter deficiencies in epilepsy. In *Neuronal Chloride Transporters in Health and Disease*. X. Tang, Ed. Elsevier, Amsterdam, Netherlands.

Oral Presentations

Rush and Helen Record Neuroscience Forum, Galveston, TX 2025
Baylor College of Medicine Neuroscience Department Annual Retreat
Mechanisms of neuronal feature selectivity in the lateral visual cortex.

Johns Hopkins University and Kennedy Krieger Institute, Baltimore, MD 2022
Pediatric Neuroscience Collaboration Seminar (virtual)
Hitting a moving target: KCC2 and pharmacoresistant seizures in the developing brain.

3rd International SYNGAP1 Conference 2021
Young Investigator Workshop (virtual)
An early forecast for the SYNGAP1 storm.

2nd International SYNGAP1 Conference, Jupiter, FL 2018
The Scripps Research Institute
Perampanel rescues behavioral state-dependent transitions on qEEG in Syngap1$^{+/-}$ *mice.*

Awards

NINDS Curing the Epilepsies Conference Travel Award 2021
CURE Epilepsy

14th European Congress on Epileptology Travel Award 2020
International League Against Epilepsy

American Epilepsy Society Annual Meeting Travel Award 2019
International League Against Epilepsy

Young Investigator Travel Award 2019
Syngap1 Foundation

Service

Ad hoc Reviewer
eLife, BRAIN, Molecular Psychiatry, Journal of Neural Engineering, Epilepsy Research, Neuropharmacology

Baylor College of Medicine Neuroscience Program Bootcamp 2022 - 2025
Lecturer and Volunteer

NINDS-AES Young Investigators Virtual Seminar 2021
Co-Organizer, Emerging neurotechnology to disrupt epilepsy

Johns Hopkins Internship in Brain Sciences 2018 - 2020
Mentor